

Assignment 9: Birth Times

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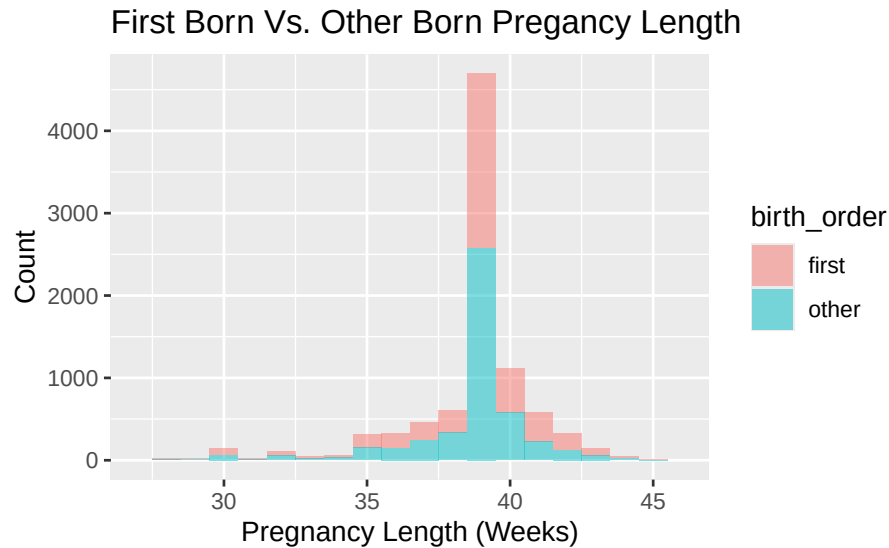
2024-10-27

Exercise 1

```
live_births <- nsfg6 %>%  
  filter(outcome == 1) %>%  
  mutate(  
    birth_order = if_else(  
      birthord == 1,  
      "first",  
      "other"  
    )  
  )  
pregnancy_length <- live_births %>%  
  select(prglngth, birth_order)
```

Exercise 2

```
pregnancy_length %>%  
  ggplot() +  
  geom_histogram(aes(x = prglngth, fill = birth_order), alpha = 0.5, binwidth = 1)+  
  xlim(27, 46)+  
  labs(title = "First Born Vs. Other Born Pregnancy Length",  
        x = "Pregnancy Length (Weeks)", y = "Count")
```



- i. Where is the mode (most common value) of the distributions?

The most common value is about 39 weeks.

- ii. By observing the visualization only, is it possible to confirm the statement that “first born children either arrive early or arrive late when compared with non-first-borns”, or should we use a hypothesis test?

No, it isn't possible. The graph of the pregnancy lengths are almost the same, visually. There should be a hypothesis test to see if the math backs up the visual data/graph.

Exercise 3

```
pregnancy_length %>%
  group_by(birth_order) %>%
  summarize(
    mean = mean(prglngth),
    median = median(prglngth),
    std_dev = sd(prglngth),
    IQR = IQR(prglngth),
    min_length = min(prglngth),
    max_length = max(prglngth)
  )
```

birth_order	mean	median	std_dev	IQR	min_length	max_length
first	38.60095	39	2.791901	1	0	48
other	38.52291	39	2.615852	0	4	50

- i. How do the different summary statistics compare between the two distributions? Does it look like there may be a notable difference between the two distributions? Explain.

For most of the summary statistics, the values are very similar. The means are very similar, 38.6 and 38.5 (first and other). The median is the same. The standard deviations are close, 2.8 and 2.6

(first and other). The Inter-Quartile Range is close as well, 1 and 0 (first and other). The minimum and maximums are; 0 and 48 for first born, 4 and 50 for other born. So, it doesn't look like there are any notable differences between first borns and other borns. All the summary values are very close, generally off by 1 or 2.

Exercise 4

The test statistic would be the difference in the means of pregnancy lengths between first borns and other borns. This will help to figure out if there's a significant difference in pregnancy lengths based on birth order.

Null Hypothesis: There's no difference in the average arrival times between first borns and other borns

Alternative Hypothesis: There is a difference in the average arrival times between first borns and other borns

The test would be two-sided because we are looking for any kind of difference between the pregnancy lengths between first borns and other borns.

Exercise 5

```

null_distribution <- pregnancy_length %>%
  specify(prglngth ~ birth_order) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 10000, type = "permute") %>%
  calculate("diff in means", order = c("first", "other"))

```

```

observed_statistic <- pregnancy_length %>%
  specify(prglngth ~ birth_order) %>%
  calculate(stat = "diff in means", order = c("first", "other"))
observed_statistic

```

stat
0.0780373

```

p_value <- null_distribution %>%
  get_p_value(obs_stat = observed_statistic, direction = "two-sided")
p_value

```

p_value
0.1694

```

if (p_value < 0.05) {
  conclusion <- "Reject the null hypothesis."
} else {
  conclusion <- "Fail to reject the null hypothesis."
}

```

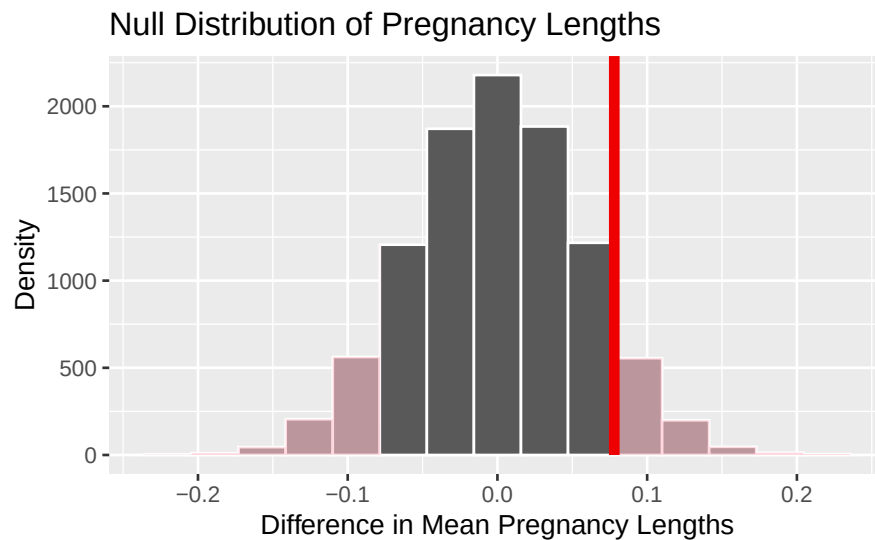
```

}
conclusion

## [1] "Fail to reject the null hypothesis."

visualization <- visualize(null_distribution) +
  shade_p_value(obs = observed_statistic, direction = "two.sided") +
  labs(title = "Null Distribution of Pregnancy Lengths",
       x = "Difference in Mean Pregnancy Lengths",
       y = "Density")
visualization

```



Exercise 6

```

birth_bootstraps <- pregnancy_length %>%
  specify(prglngth ~ birth_order) %>%
  generate(10000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("first", "other"))

bootstrap_ci <- birth_bootstraps %>%
  get_confidence_interval(level = 0.95)
bootstrap_ci

```

lower_ci	upper_ci
-0.0345438	0.1901824

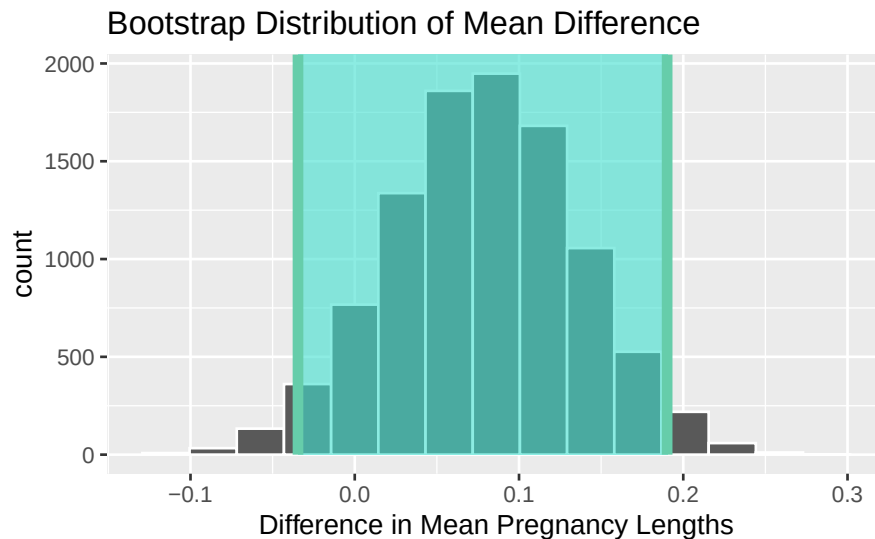
- i. Does the observed statistic fall within the range of the 95% confidence interval?

Since the observed statistic (0.0780) falls between the lower bound (-0.0327) and the upper bound (0.1905) of the 95% confidence interval, the observed statistic does fall within the range of the 95% confidence interval.

```

birth_bootstraps %>%
  visualize() +
  shade_confidence_interval(bootstrap_ci) +
  labs(title = "Bootstrap Distribution of Mean Difference",
       x = "Difference in Mean Pregnancy Lengths")

```



Exercise 7

```

bootstrap_results <- cohens_d_bootstrap(data = pregnancy_length,
                                       model = prglngth ~ birth_order)
bootstrap_report(bootstrap_results)

```

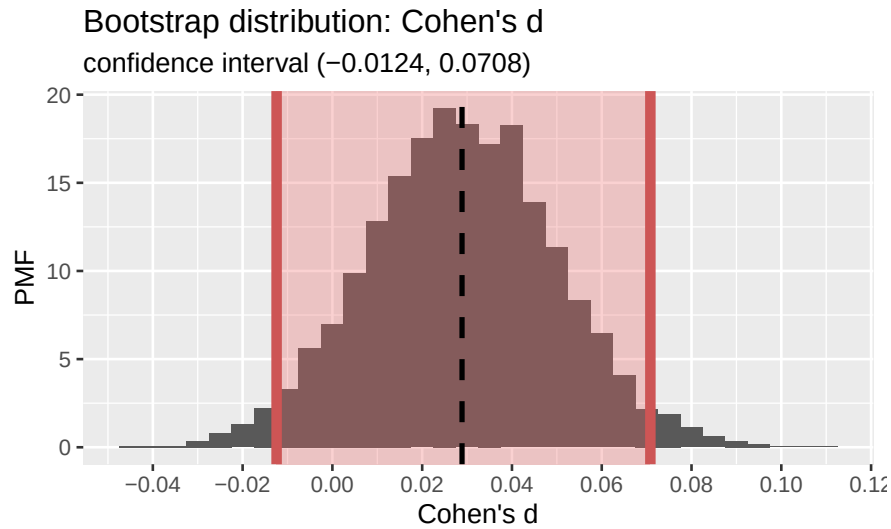
```

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot::boot.ci(boot.out = cohens_d_bootstrap_sim, type = c("perc"))
##
## Intervals :
## Level      Percentile
## 95%      (-0.0124,  0.0708 )
## Calculations and Intervals on Original Scale
##
## Response variable
## prglngth
##
## Explanatory variable
## birth_order
##
## Explanatory category with larger mean
## first
##

```

```
## Explanatory category with smaller mean
## other
##
## Cohen's d observed value
## 0.0288791
```

```
plot_ci(bootstrap_results)
```



The effect size for the difference in pregnancy lengths between first births and other births is very small. This means that the difference in pregnancy lengths doesn't change based on the birth order.

Academic integrity statement

Online tools and guidance from SLACK was used to complete this assignment.

- infer.netlify.app/reference/hypothesize
- chatGPT